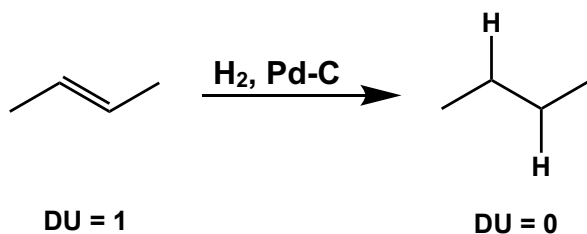


Activity 2: ¹H NMR Workshop 1 Practice Problems

Degree of Unsaturation (DU): Describes the number of pi bonds and rings in a molecule and is equivalent to half the number of hydrogens required to fully “saturate” a molecule

- For example, an alkene has a DU = 1, and can be hydrogenated to add two hydrogens



Degree of Unsaturation can be described by the following equations:

$$DU = \frac{2C + 2 + N - H - X}{2}$$

$$DU = \#Rings + \#pi \text{ bonds}$$

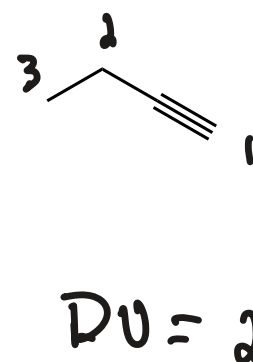
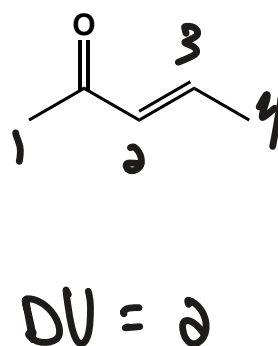
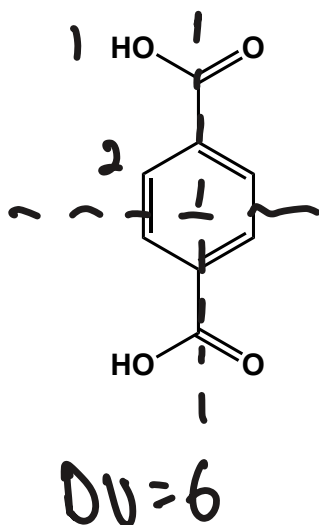
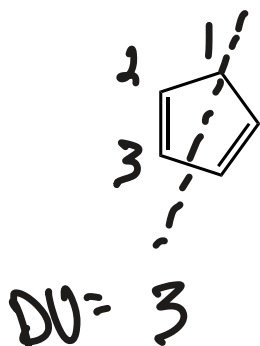
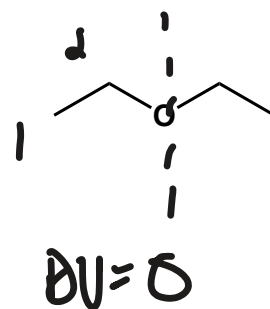
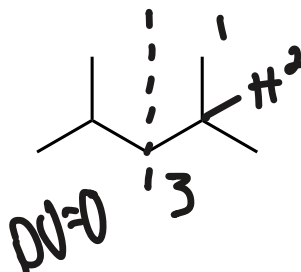
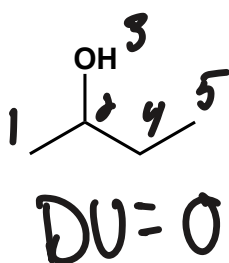
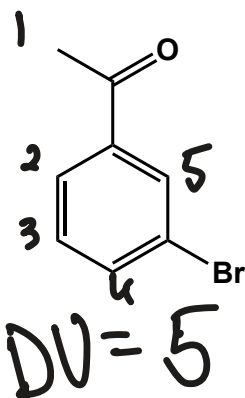
The **number of signals** observed for a given molecule relates to the number of protons in *unique chemical environments*.

- Check for symmetry!
- Remember free rotation of bonds

The **integration** of a signal, or the area under the curve of a peak, is directly *proportional* to the number of hydrogens represented by that signal

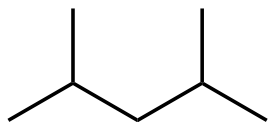
- Integration is a ratio!

1. Assign a Degree of Unsaturation and predict the number of signals expected in the for the following molecules.

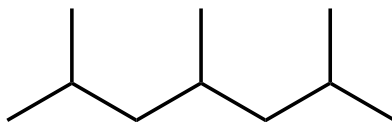


Activity 2: ^1H NMR Workshop 1 Practice Problems

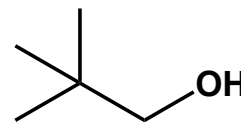
2. What is the ratio of methylene ($-\text{CH}_2$) to methyl ($-\text{CH}_3$) protons in the following molecules?



6:1



15:4

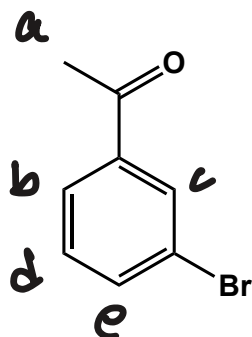


9:2

Chemical Shift (δ) is the position of the signal on the x-axis of an NMR spectrum. Chemical shift is related to the unique chemical environment of a proton and is influenced by the electronics of the molecule.

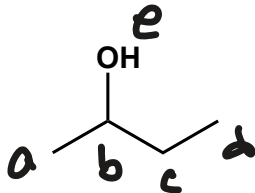
- See NMR characteristic shift table for *approximate* chemical shift ranges
- Consider inductive and resonance effects

3. Identify approximate chemical shift ranges for each unique chemical environment of the following molecules.



a: 2-2.7

b-e: 6.5-8

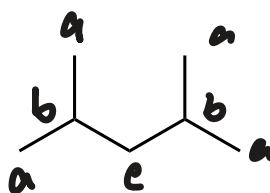


a, d: ~0.9

b: 3.3-4

c: ~1.3

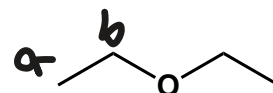
e: 0.5-5



a: ~0.9

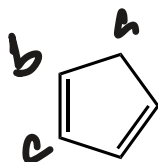
b: ~1.5

c: ~1.3



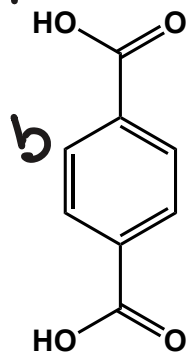
a: 0.9

b: 3.3-4



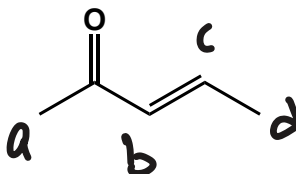
a: ~1.7

b-c: 4.5-6



a: 10-12

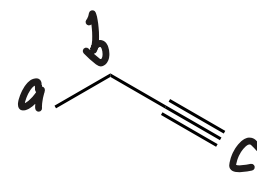
b: 6.5-8



a: 2-2.7

b-c: 4.5-6

d: ~1.7



a: ~0.9

b: ~1.5 2° or allylic

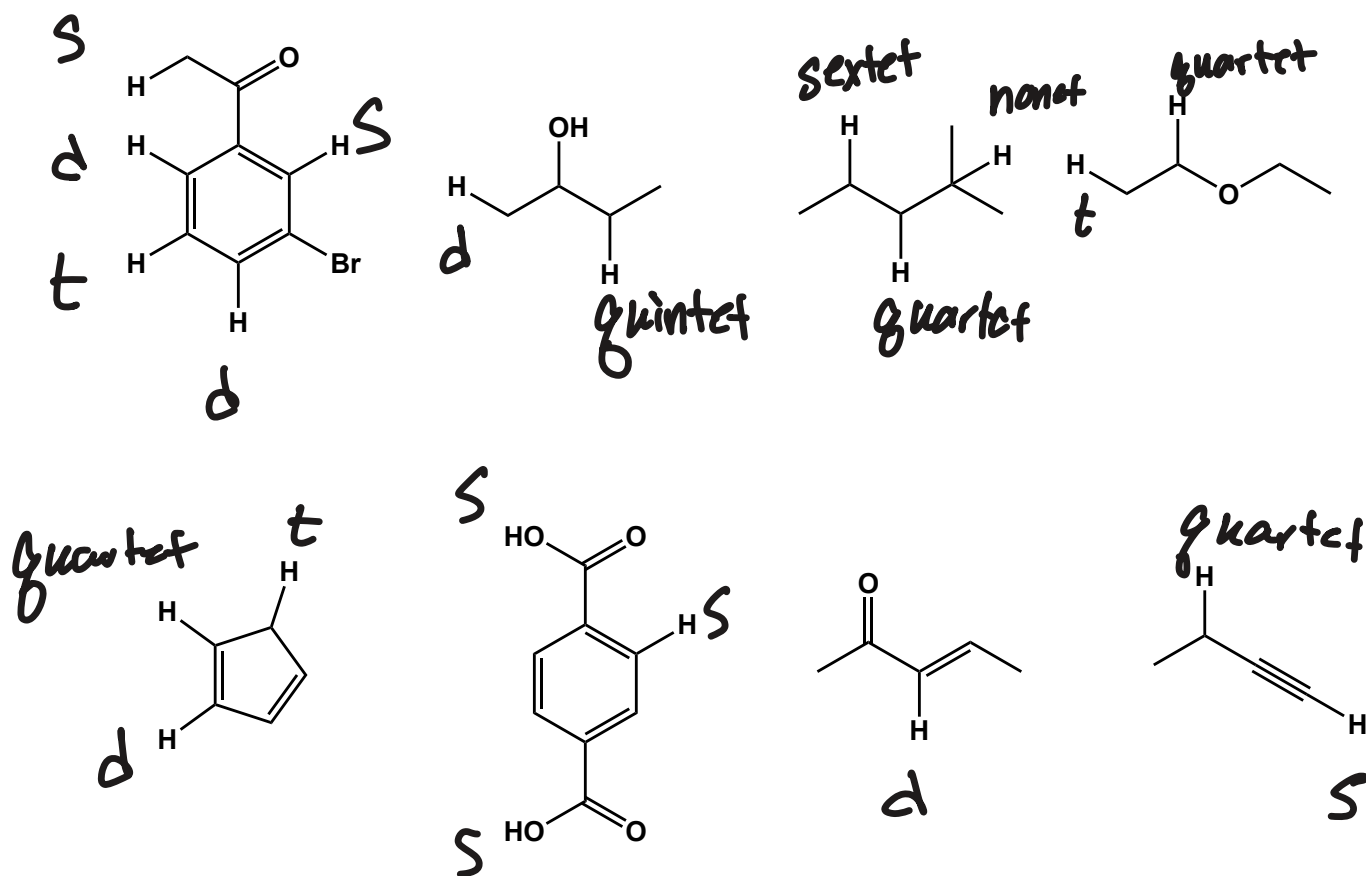
c: 2-3

Activity 2: ^1H NMR Workshop 1 Practice Problems

Splitting of an NMR signal occurs when there are neighboring protons within three bonds. This is called coupling.

- Protons cannot couple with equivalent protons ("you can't couple with yourself")
- Coupling only occurs within three bonds
- Coupling does not occur with protons bonded to oxygen or hydrogen; O-H and N-H always appear as singlets
- Follow the "n+1" rule, where n = number of neighboring hydrogens

4. Identify the splitting patterns for all labeled hydrogens. Note that not all hydrogens are explicitly shown.



Problem solving strategy:

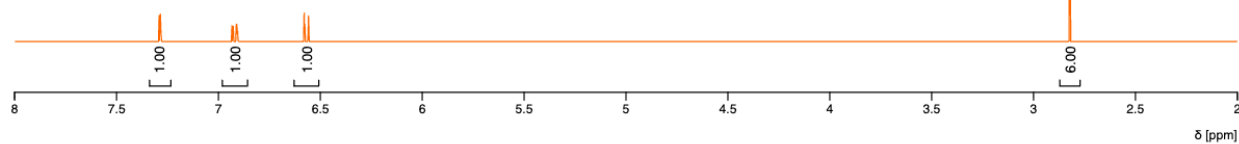
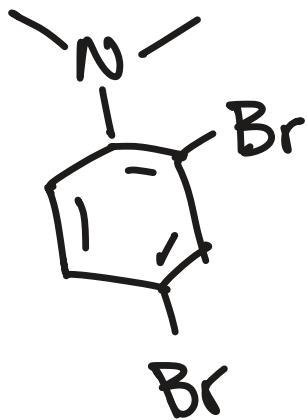
Most problems that require you to use spectral data to identify a chemical structure can be solved using one or more of the above topics. Generally, you can narrow down your options by using the following strategy:

1. Calculate degree of unsaturation (tells us the number of rings + the number of pi bonds)
2. Identify the number of signals in the spectrum
3. Look for approximate chemical shift ranges to identify functional groups
4. Check for splitting patterns to identify how to arrange atoms on the molecule (i.e. are there neighboring protons, how many?)

Activity 2: ^1H NMR Workshop 1 Practice Problems

5. Identify the molecules represented by the following NMR Spectra:

$\text{C}_8\text{H}_9\text{Br}_2\text{N}$



$\text{C}_4\text{H}_8\text{O}_2$

